

## CSCI 315 – Assignment 4 Writeup

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\*To keep this writeup under two pages as per instructions, we only include one confusion matrix per table

### Adjusting # of Epochs

| # of Epochs | # of Features | Learning Rate | Test Loss | Test Accuracy |
|-------------|---------------|---------------|-----------|---------------|
| 20          | 256           | 0.001         | 0.0062374 | 88.99284%     |
| 40          | 256           | 0.001         | 0.0049846 | 90.66481%     |
| 60          | 256           | 0.001         | 0.0044669 | 91.52070%     |

Confusion Matrix for 40 Epochs

| Actual Digit | Classification Output |      |     |     |     |     |     |     |     |     |
|--------------|-----------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|
|              | 0                     | 1    | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   |
| 0            | 955                   | 0    | 4   | 2   | 0   | 7   | 8   | 1   | 3   | 0   |
| 1            | 0                     | 1108 | 1   | 4   | 1   | 1   | 4   | 1   | 15  | 0   |
| 2            | 11                    | 6    | 904 | 17  | 16  | 1   | 14  | 15  | 41  | 7   |
| 3            | 3                     | 1    | 20  | 910 | 0   | 28  | 3   | 15  | 23  | 7   |
| 4            | 1                     | 3    | 6   | 1   | 908 | 1   | 11  | 2   | 8   | 41  |
| 5            | 12                    | 3    | 5   | 42  | 12  | 752 | 17  | 10  | 31  | 8   |
| 6            | 15                    | 3    | 4   | 1   | 17  | 16  | 899 | 1   | 2   | 0   |
| 7            | 4                     | 15   | 26  | 5   | 9   | 0   | 0   | 934 | 4   | 31  |
| 8            | 8                     | 8    | 8   | 29  | 10  | 26  | 12  | 13  | 848 | 12  |
| 9            | 11                    | 7    | 4   | 12  | 42  | 14  | 0   | 21  | 6   | 892 |

It seems that the number of epochs has a considerable amount of impact on accuracy.

### Adjusting # of Features

| # of Epochs* | # of Features | Learning Rate | Training Time (s) | Test Loss | Test Accuracy |
|--------------|---------------|---------------|-------------------|-----------|---------------|
| 102          | 64            | 0.001         | 125.008           | 0.0039354 | 92.41640%     |
| 92           | 256           | 0.001         | 127.656           | 0.0039176 | 92.44626%     |
| 86           | 576           | 0.001         | 135.463           | 0.0039095 | 92.52588%     |

\*model was trained to learn until train\_loss < 0.26 (arbitrarily, but empirically determined)

Confusion Matrix for 16 Features

| Actual Digit | Classification Output |      |     |     |     |     |     |     |     |     |
|--------------|-----------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|
|              | 0                     | 1    | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   |
| 0            | 961                   | 0    | 3   | 2   | 0   | 3   | 8   | 1   | 2   | 0   |
| 1            | 0                     | 1109 | 3   | 2   | 1   | 2   | 4   | 1   | 13  | 0   |
| 2            | 9                     | 6    | 933 | 16  | 9   | 0   | 13  | 13  | 29  | 4   |
| 3            | 3                     | 1    | 18  | 925 | 0   | 22  | 2   | 12  | 18  | 9   |
| 4            | 2                     | 2    | 4   | 1   | 922 | 0   | 11  | 2   | 5   | 33  |
| 5            | 10                    | 3    | 7   | 35  | 10  | 781 | 15  | 5   | 19  | 7   |
| 6            | 10                    | 3    | 4   | 1   | 13  | 13  | 911 | 1   | 2   | 0   |
| 7            | 2                     | 10   | 25  | 7   | 5   | 0   | 0   | 953 | 3   | 23  |
| 8            | 7                     | 6    | 5   | 22  | 9   | 16  | 12  | 11  | 878 | 8   |
| 9            | 10                    | 7    | 1   | 11  | 36  | 8   | 1   | 14  | 5   | 916 |

It seems that using more features barely improves accuracy, contrary to our beliefs on the tradeoff of features.

### Adjusting Learning Rate Eta

| # of Epochs* | # of Features** | Learning Rate | Training Time (s) | Test Loss | Test Accuracy |
|--------------|-----------------|---------------|-------------------|-----------|---------------|
| 92           | 256             | 0.001         | 127.656           | 0.0039176 | 92.44626%     |
| 10           | 256             | 0.01          | 13.121            | 0.0038087 | 92.57564%     |
| 2            | 256             | 0.1           | 2.758             | 0.0028160 | 94.33718%     |

*Confusion Matrix for Learning Rate = 0.01*

| Actual Digit | Classification Output |      |     |     |     |     |     |     |     |     |
|--------------|-----------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|
|              | 0                     | 1    | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   |
| 0            | 959                   | 0    | 3   | 2   | 0   | 4   | 9   | 1   | 2   | 0   |
| 1            | 0                     | 1111 | 3   | 2   | 1   | 3   | 4   | 1   | 10  | 0   |
| 2            | 6                     | 6    | 945 | 15  | 10  | 3   | 12  | 13  | 19  | 3   |
| 3            | 2                     | 1    | 20  | 927 | 0   | 25  | 2   | 14  | 13  | 6   |
| 4            | 1                     | 3    | 4   | 1   | 930 | 0   | 12  | 2   | 4   | 25  |
| 5            | 10                    | 3    | 6   | 33  | 10  | 786 | 16  | 5   | 16  | 7   |
| 6            | 8                     | 3    | 4   | 1   | 12  | 12  | 915 | 1   | 2   | 0   |
| 7            | 2                     | 10   | 29  | 3   | 6   | 0   | 0   | 957 | 2   | 19  |
| 8            | 6                     | 8    | 6   | 26  | 10  | 21  | 12  | 13  | 864 | 8   |
| 9            | 9                     | 8    | 1   | 11  | 42  | 10  | 1   | 15  | 4   | 908 |

It is surprising that having a learning rate of 0.1 improves accuracy and significantly reduces training time.

### SLP: Adjusting # of Epochs

| # of Epochs* | Learning Rate | Training Time (s) | Test Loss | Test Accuracy |
|--------------|---------------|-------------------|-----------|---------------|
| 40           | 0.001         | 45.035            | 0.0062041 | 89.24164%     |
| 60           | 0.001         | 70.222            | 0.0056985 | 89.80891%     |
| 120          | 0.001         | 151.245           | 0.0050741 | 90.66481%     |
| 240          | 0.001         | 290.349           | 0.0046639 | 91.46099%     |

*Confusion Matrix for # of Epochs = 40*

| Actual Digit | Classification Output |      |     |     |     |     |     |     |     |     |
|--------------|-----------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|
|              | 0                     | 1    | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   |
| 0            | 960                   | 0    | 2   | 3   | 0   | 1   | 6   | 1   | 7   | 0   |
| 1            | 0                     | 1103 | 2   | 4   | 1   | 2   | 4   | 0   | 19  | 0   |
| 2            | 13                    | 8    | 876 | 22  | 17  | 0   | 20  | 22  | 46  | 8   |
| 3            | 6                     | 1    | 17  | 895 | 1   | 33  | 6   | 16  | 23  | 12  |
| 4            | 1                     | 6    | 5   | 0   | 899 | 1   | 12  | 1   | 8   | 49  |
| 5            | 16                    | 7    | 6   | 48  | 16  | 720 | 16  | 10  | 43  | 10  |
| 6            | 16                    | 3    | 7   | 2   | 12  | 18  | 894 | 1   | 5   | 0   |
| 7            | 2                     | 20   | 30  | 3   | 12  | 0   | 1   | 912 | 5   | 43  |
| 8            | 10                    | 9    | 12  | 30  | 10  | 26  | 13  | 13  | 836 | 15  |
| 9            | 13                    | 8    | 10  | 11  | 48  | 15  | 0   | 26  | 6   | 872 |

*Confusion Matrix for # of Epochs = 240*

| Actual Digit | Classification Output |      |     |     |     |     |     |     |     |     |
|--------------|-----------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|
|              | 0                     | 1    | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   |
| 0            | 962                   | 0    | 2   | 2   | 0   | 3   | 8   | 1   | 2   | 0   |
| 1            | 0                     | 1105 | 2   | 2   | 0   | 2   | 4   | 2   | 18  | 0   |
| 2            | 9                     | 7    | 911 | 16  | 15  | 1   | 13  | 13  | 40  | 7   |
| 3            | 3                     | 1    | 21  | 919 | 0   | 26  | 2   | 11  | 17  | 10  |
| 4            | 1                     | 1    | 5   | 1   | 914 | 0   | 12  | 2   | 8   | 38  |
| 5            | 10                    | 4    | 3   | 35  | 9   | 770 | 17  | 7   | 29  | 8   |
| 6            | 12                    | 3    | 3   | 2   | 11  | 13  | 911 | 1   | 2   | 0   |
| 7            | 3                     | 13   | 22  | 7   | 8   | 0   | 0   | 939 | 3   | 33  |
| 8            | 6                     | 9    | 9   | 23  | 9   | 24  | 13  | 13  | 859 | 9   |
| 9            | 11                    | 7    | 3   | 11  | 38  | 8   | 0   | 24  | 7   | 900 |

It appears that with enough epochs, we still achieve good results with a single layer perceptron.